

Numerical Differentiation



Recall

$$\frac{df}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Taylor Theorem:

$$f(x+h) = f(x) + f'(x)h + \frac{f^{(2)}(x)h^2}{2!} + \frac{f^{(3)}(x)h^3}{3!} + O(h^4)$$

$$E = O(h^n) \Rightarrow \exists \text{ real, finite } C, \text{ such that: } |E| \leq C|h|^n$$

E is of order $h^n \Rightarrow E$ is approaching zero at rate similar to h^n

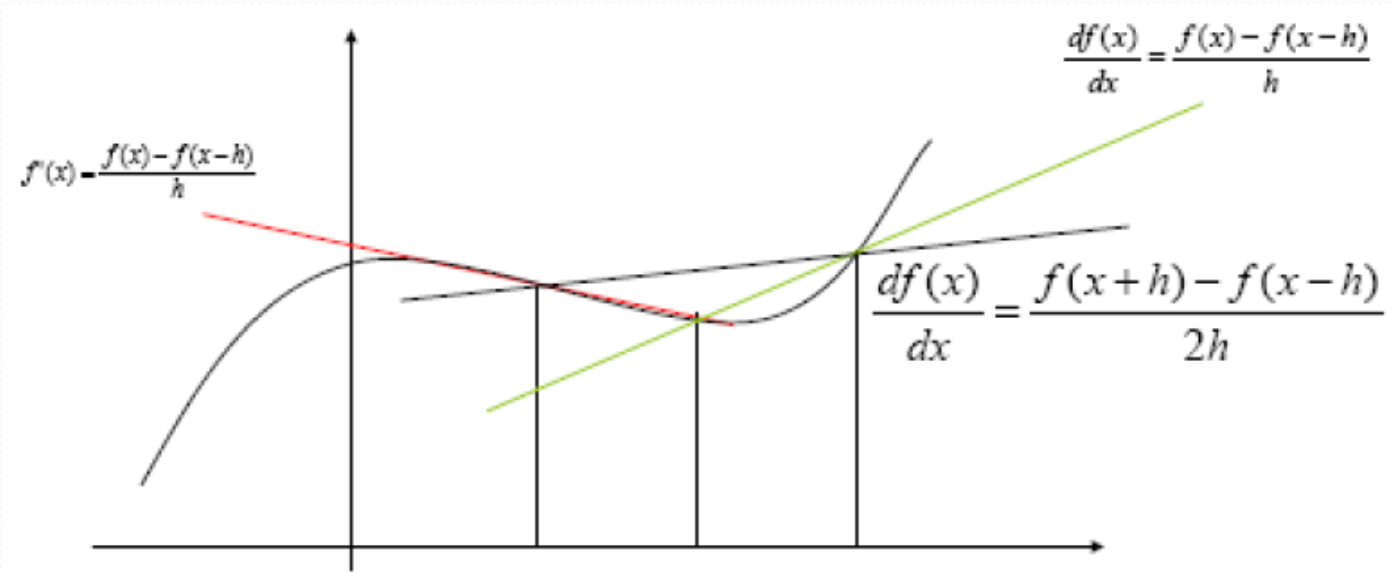
Three Formula

Forward Difference: $\frac{df(x)}{dx} = \frac{f(x+h) - f(x)}{h}$

Backward Difference: $\frac{df(x)}{dx} = \frac{f(x) - f(x-h)}{h}$

Central Difference: $\frac{df(x)}{dx} = \frac{f(x+h) - f(x-h)}{2h}$

The Three Formulas



Newton's forward difference differential formula

- Assume that $(n+1)$ points (x_i, y_i) , $i = 1, 2, \dots, n$ for any function $y = f(x)$ are given. The values of x being equally spaced, i.e., $x_i = x_0 + ih$, $i = 1, 2, \dots, n$
- To find the derivative of $y = f(x)$ at a point x_0 .
- From the Newton's forward difference formula, we know

$$\phi(x) = y_0 + p \Delta y_0 + \frac{p(p-1)}{2!} \Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!} \Delta^3 y_0 + \dots + \frac{p(p-1)\dots(p-n+1)}{n!} \Delta^n y_0$$

Cont.

where

$$x - x_0 = ph$$

$$\Rightarrow \frac{dp}{dx} = \frac{1}{h}$$

$$\Rightarrow \frac{dy}{dx} = \frac{dp}{dx} \times \frac{dy}{dp} = \frac{1}{h} \times \frac{dy}{dp}$$

$$f'(x) = \frac{dy}{dx}$$

$$= \frac{1}{h} \left[\Delta y_0 + \frac{(2p-1)}{2!} \Delta^2 y_0 + \frac{3p^2 - 6p + 2}{3!} \Delta^3 y_0 + \frac{4p^3 - 18p^2 + 22p - 6}{4!} \Delta^4 y_0 + \dots \right]$$



Cont

Similarly

$$f'(x) = \frac{dy}{dx} \\ = \frac{1}{h} \left[\Delta y_0 + \frac{(2p-1)}{2!} \Delta^2 y_0 + \frac{3p^2-6p+2}{3!} \Delta^3 y_0 + \frac{4p^3-18p^2+22p-6}{4!} \Delta^4 y_0 + \dots \right]$$

$$f''(x) = \frac{dy}{dx} = \frac{1}{h^2} \left[\Delta^2 y_0 + (p-1) \Delta^3 y_0 + \frac{12p^2-36p+22}{4!} \Delta^4 y_0 + \dots \right]$$

$$f'''(x) = \frac{dy}{dx} = \frac{1}{h^3} \left[\Delta^3 y_0 + \frac{24p-36}{4!} \Delta^4 y_0 + \dots \right]$$

Cont.

• Now $x = x_0 \Rightarrow p = 0$

$$f'(x_0) = \frac{dy}{dx}_{x=x_0} = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right]$$

$$f''(x_0) = \frac{dy}{dx}_{x=x_0} = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 + \dots \right]$$

$$f'''(x_0) = \frac{dy}{dx}_{x=x_0} = \frac{1}{h^3} \left[\Delta^3 y_0 - \frac{3}{2} \Delta^4 y_0 + \dots \right]$$

Backward difference differential formula

$$\begin{aligned} f'(x) &= \frac{dy}{dx} \\ &= \frac{1}{h} \left[\Delta y_n + \frac{(2p+1)}{2!} \Delta^2 y_n + \frac{3p^2+6p+2}{3!} \Delta^3 y_n + \frac{4p^3+18p^2+22p+6}{4!} \Delta^4 y_n + \dots \right] \end{aligned}$$

$$f''(x) = \frac{d^2y}{dx^2} = \frac{1}{h^2} \left[\Delta^2 y_n + (p+1) \Delta^3 y_n + \frac{12p^2+36p+22}{4!} \Delta^4 y_n + \dots \right]$$

$$f'''(x) = \frac{d^3y}{dx^3} = \frac{1}{h^3} \left[\Delta^3 y_n + \frac{2p+3}{2} \Delta^4 y_n + \dots \right]$$

Backward_{cont}

- Now

$$x = x_n \Rightarrow p = 0$$

$$f'(x_n) = \frac{dy}{dx}_{x=x_n} = \frac{1}{h} \left[\Delta y_n + \frac{1}{2} \Delta^2 y_n + \frac{1}{3} \Delta^3 y_n + \frac{1}{4} \Delta^4 y_n + \dots \right]$$

$$f''(x_n) = \frac{d^2y}{dx^2}_{x=x_n} = \frac{1}{h^2} \left[\Delta^2 y_n + \Delta^3 y_n + \frac{11}{12} \Delta^4 y_n + \dots \right]$$

$$f'''(x_n) = \frac{d^3y}{dx^3}_{x=x_n} = \frac{1}{h^3} \left[\Delta^3 y_0 + \frac{3}{2} \Delta^4 y_n + \dots \right]$$

Estimate the velocity and the acceleration of a moving particle at $t = 0.3$, whose distance along a straight line at various times are as follows:

t(sec)	0.1	0.2	0.3	0.4	0.5
x(m)	31.62	32.87	33.64	33.95	33.81

Ans: 5.3750 m/s, -45.9167 m/s²

Find the best possible value of $y'(93)$ from the following table:

x	60	75	90	105	120
y	28.2	38.2	43.2	40.9	37.0

Ans: -0.03627